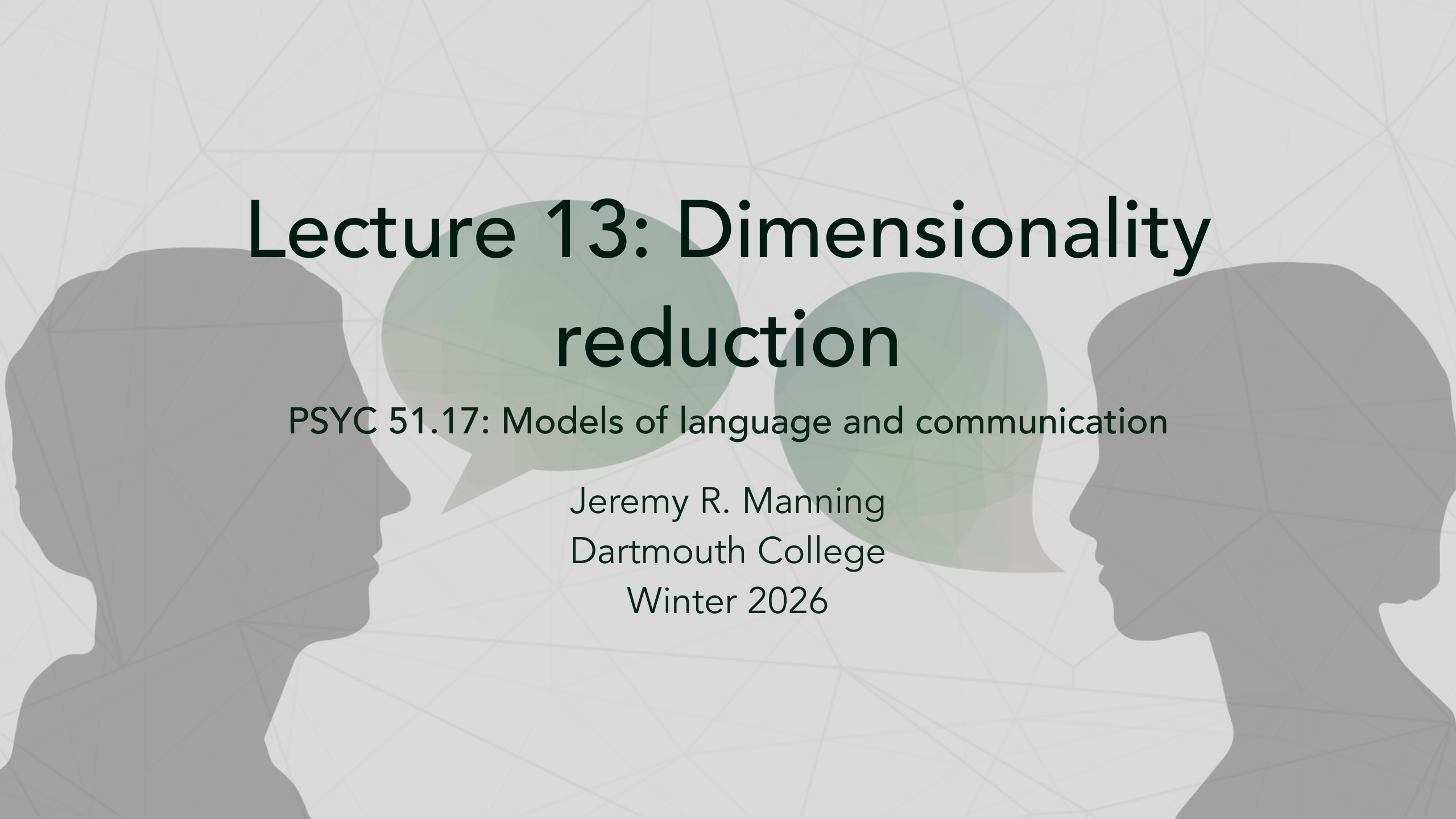




Lecture 13: Dimensionality reduction

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will

1. Understand **why** we need dimensionality reduction for embeddings
2. Apply **PCA** for fast, linear projection
3. Apply **t-SNE** for local structure preservation
4. Apply **UMAP** for scalable, global+local structure
5. Know **when to use** each technique

The goal

Reduce from d dimensions (e.g., 300-768) to 2-3 dimensions while preserving meaningful structure for visualization.

Why dimensionality reduction?

The curse of dimensionality

High-dimensional spaces behave counter-intuitively: all points tend to be equidistant!

- **Word2Vec / GloVe:** 300 dimensions
- **BERT:** 768 dimensions
- **GPT-3:** 12,288 dimensions

Applications

- **Visualization:** 2D/3D plots to explore semantic structure
- **Quality assurance:** Verify embeddings capture expected relationships
- **Preprocessing:** Reduce noise before clustering
- **Interpretation:** Understand relationships between words/documents

PCA: Principal Component Analysis

The classic linear method

Finds directions of **maximum variance** and projects data onto these orthogonal "principal components."

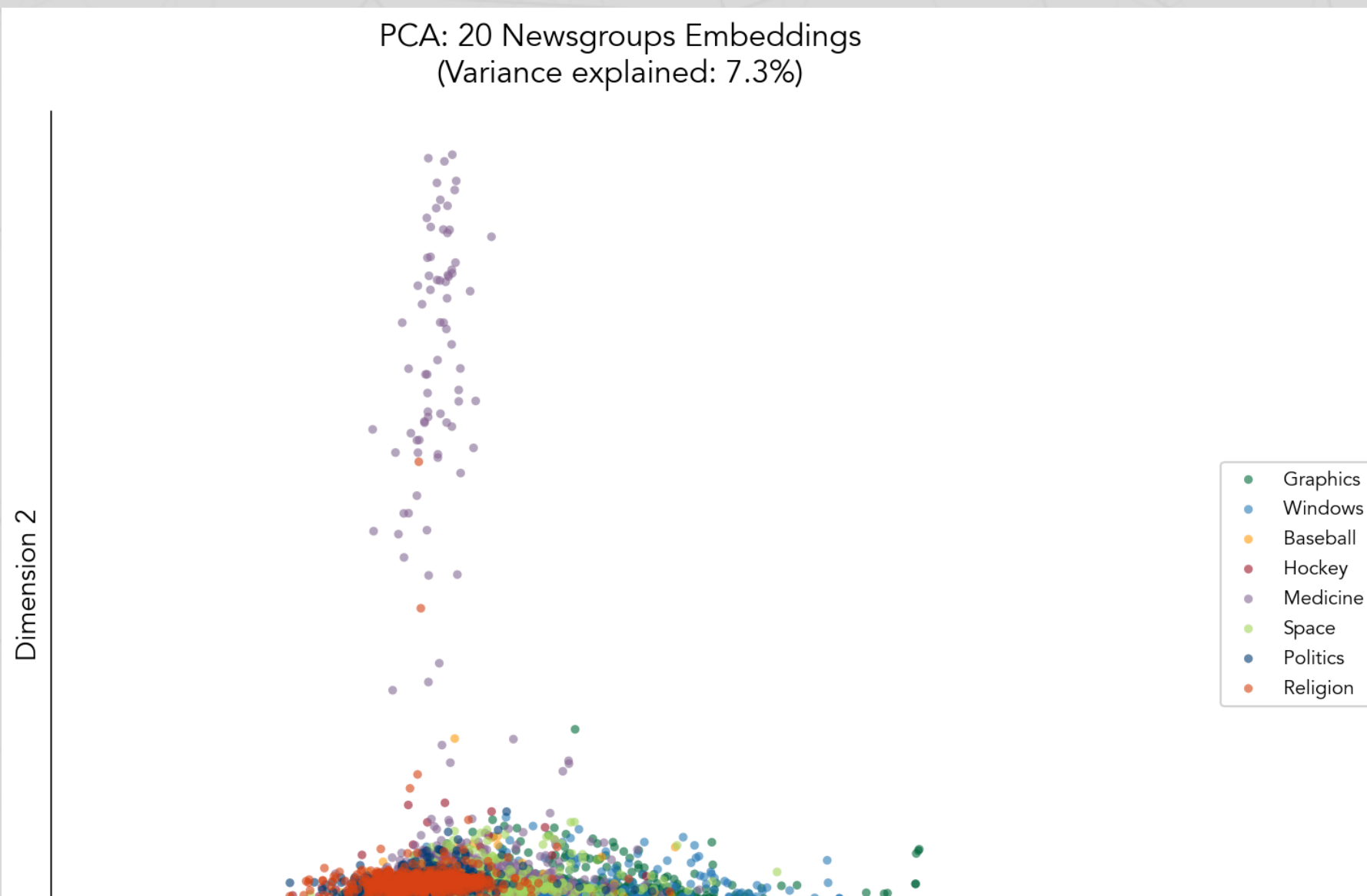
Properties

- **Linear transformation** — preserves global structure
- **Deterministic** — same result every time
- **Fast** — efficient computation via SVD
- **Interpretable** — PCs ranked by variance explained

When to use PCA

Quick first exploration, preprocessing for other methods (like t-SNE/UMAP), or when computational speed is critical.

PCA: 20 Newsgroups visualization



t-SNE: t-Distributed Stochastic Neighbor Embedding

Further reading

van der Maaten & Hinton (2008, *JMLR*) Visualizing Data using t-SNE.

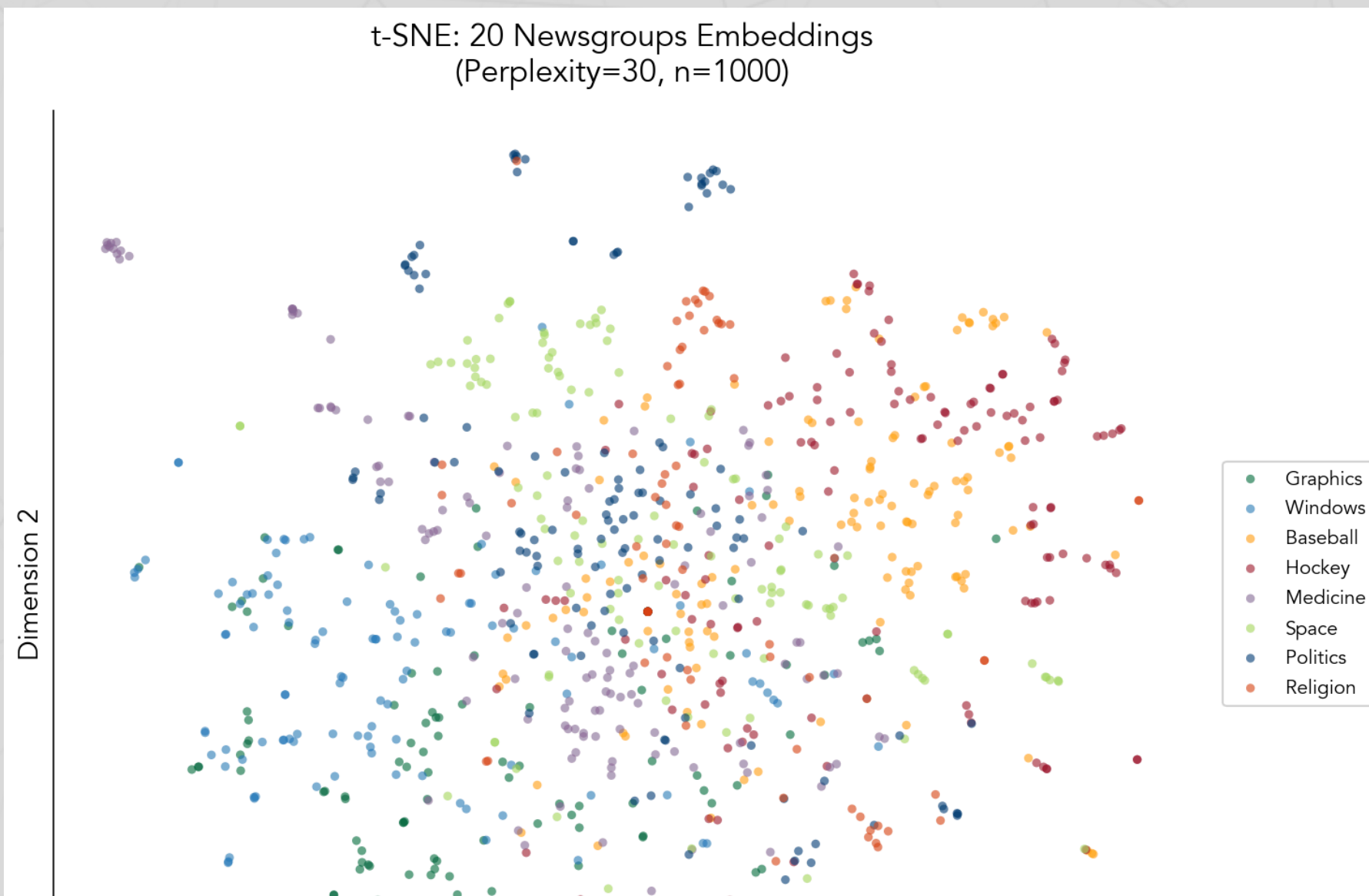
Non-linear visualization

Models similarity as probability distributions. Preserves **local structure** — neighbors stay neighbors.

Caveats

- **Slow:** $O(N^2)$ complexity
- **Non-deterministic:** Different seeds → different layouts
- **Cluster sizes are arbitrary** — don't interpret "bigger" as "more variance"
- **Distances between clusters are meaningless**

t-SNE: 20 Newsgroups visualization



UMAP: Uniform Manifold Approximation & Projection

Further reading

McInnes et al. (2018, [arXiv](#)) UMAP: Uniform Manifold Approximation and Projection.

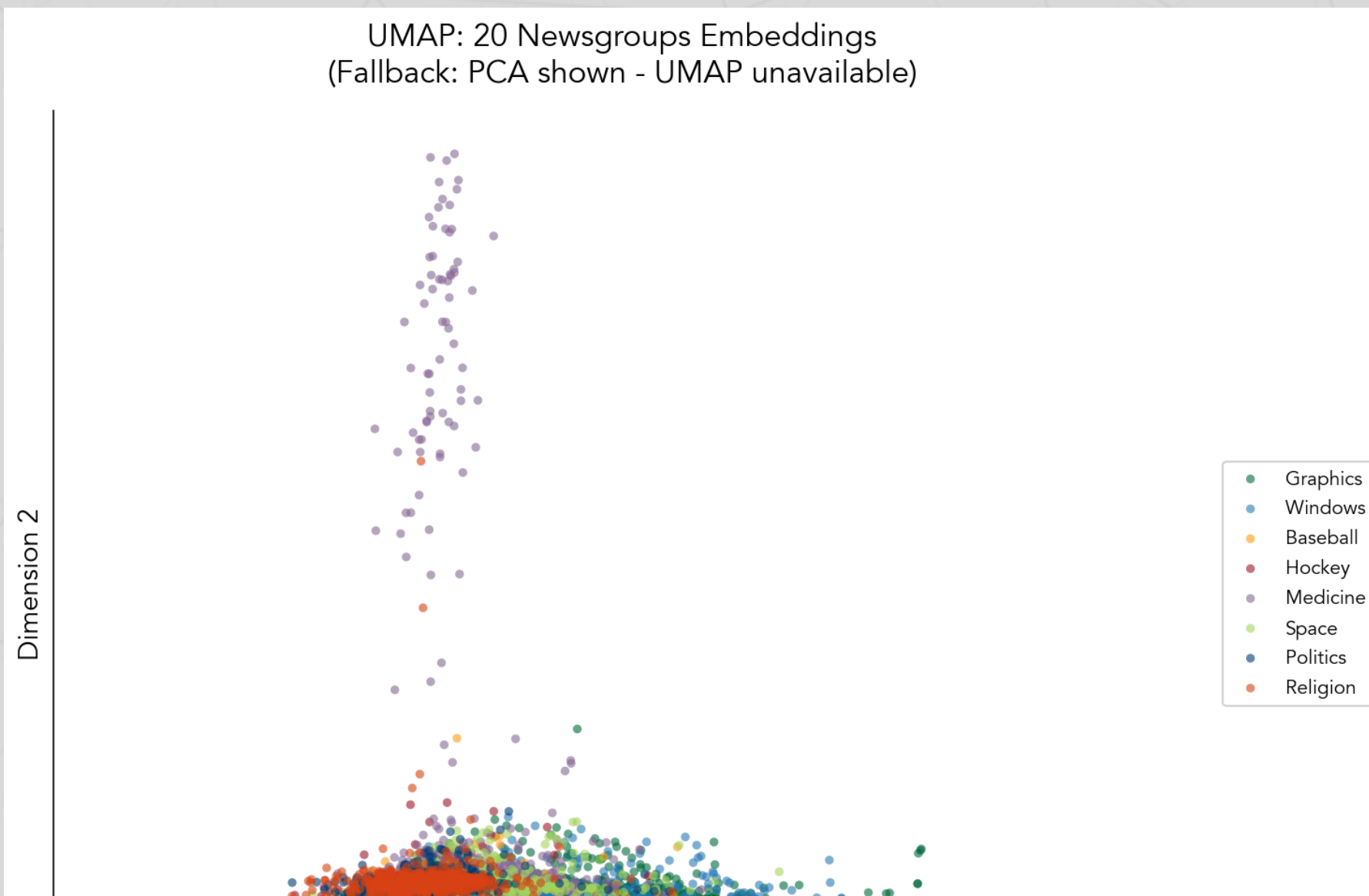
The modern standard (2018)

Theoretically grounded in topology. Preserves **both local AND global** structure.

Key advantages

- **Fast:** $O(N \log N)$ complexity
- **Scalable:** Handles millions of points
- **Out-of-sample:** Can transform new points without refitting
- **Global structure:** Keeps related clusters near each other

UMAP: 20 Newsgroups visualization



PCA vs. t-SNE vs. UMAP

Feature	PCA	t-SNE	UMAP
Speed	Very Fast	Slow	Fast
Scalability	Excellent	Poor (<10k)	Excellent
Global structure	Yes	No	Yes
Local structure	Partial	Yes	Yes
Embedding			

Visualization best practices

Preparation

- **Standardize:** Use `StandardScaler` for PCA
- **Sample:** For very large datasets, subsample or use PCA first
- **Iterate:** Try multiple hyperparameters

Presentation

- **Color intelligently:** By semantic category, cluster, or feature
- **Interactive:** Use Plotly, datamapplot, or Bokeh for hover tooltips
- **Annotate selectively:** Label representative points, not every one

What you CANNOT conclude

- Exact distances or true high-D geometry
- Quantitative relationships between distant clusters
- Absolute cluster sizes (especially in t-SNE)

Summary

What we learned

1. **Curse of Dimensionality:** High-D space is counter-intuitive; reduction essential for visualization
2. **PCA:** Linear, fast, global focus. Best for noise removal and preprocessing.
3. **t-SNE:** Non-linear, local focus. Great for clusters but slow.
4. **UMAP:** Fast, scalable, preserves local and global structure. The modern default.
5. **Workflow:** PCA (50D) → UMAP (2D) is the gold standard.

Try it yourself!



X-hour Notebook — Interactive demo with BERTopic and datamapplot

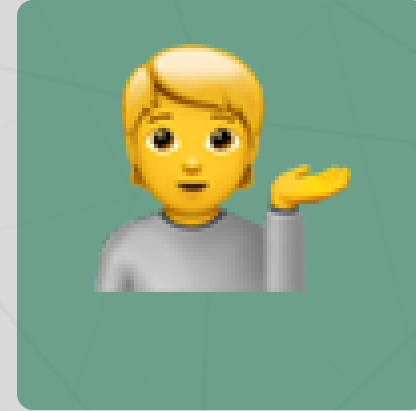
Questions?



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Come to office hours

Up next

Friday: Cognitive models of semantic representation — how do humans represent meaning?